

Discriminating and co-discriminating groups

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1 Introduction

1.1 Separation conditions

This paper is concerned with proving that a number of groups satisfy certain separation conditions. In order to explain what these conditions are we need to recall some definitions from [5] (see also [4]).

Definition 1 *Let H be a group, $\mathcal{S}$ a non-empty set of groups and $\mathcal{F}$ a family of homomorphisms of H into the groups in $\mathcal{S}$. We term $\mathcal{F}$ a separating family if for each $h \in H$, $h \neq 1$, there exists $\phi \in \mathcal{F}$ such that $\phi(h) \neq 1$. In this event we also say that $\mathcal{S}$ separates H or that H is residually $\mathcal{S}$.*

If $\mathcal{S}$ consists of the singleton G , then we say that G separates H or that H is residually G .

Definition 2 *Let H be a group, $\mathcal{S}$ a non-empty set of groups and $\mathcal{F}$ a family of homomorphisms of H into the groups in $\mathcal{S}$. We term $\mathcal{F}$ a discriminating family if for each finite subset Y of non-trivial elements of H , there exists a homomorphism $\phi \in \mathcal{F}$ such that $\phi(h) \neq 1$ for every $h \in Y$ (we say that ϕ discriminates the finite set Y). In this event we say that $\mathcal{S}$ discriminates H .*

If $\mathcal{S}$ consists of a singleton G we say that G discriminates H , or that H is fully residually G or that H is ω -residually G .

These notions of separation and discrimination play a role in several areas of group theory; for example, in the theory of varieties of groups [17], in algorithmic group theory [14] and in, what we have termed, algebraic geometry over groups [5]. Our concern here is with two related notions. However,

before we turn our attention to them, it may be worth noting that there is a distinct difference between the notion of separation and that of discrimination. The interested reader will find it instructive to reflect on the notion of a root property discussed in the paper [7], the results in [2], [3], [6] and the book [17].

Our concern here will be with two notions related to separation and discrimination.

1.2 Discriminating groups

The first of these is that of a *discriminating* group. We term a group G *discriminating* if every group separated by G is discriminated by G . It is not easy to determine which groups are discriminating, although it is not hard to find a characterization of such groups. Indeed *a group G is discriminating if and only if its direct square $G \times G$ is discriminated by G* (see section 2). We concentrate here on the task of classifying those abelian groups which are discriminating. It turns out that every torsion-free abelian group is discriminating. Our main result here, Theorem A, involves torsion abelian groups. We refer the reader to the books (see [9], [11]) for a comprehensive discussion of abelian groups and the notions described below. In order to formulate Theorem A we need to digress a little. Let then A be a torsion abelian group. Given an integer n , we define

$$A[n] = \{a \in A \mid a^n = 1\}.$$

Clearly, $A[n]$ is a subgroup of A . Observe that $A[p^{k-1}] \leq A[p^k]$, for every prime p and every positive integer k . Moreover $A[p^k]/A[p^{k-1}]$ is of exponent p and can therefore be viewed as a vector space over the field of p elements. We denote the dimension of a vector space V by $\dim(V)$ and define

$$\alpha(A, p, k) = \dim(A[p^k]/A[p^{k-1}]).$$

Consider next the Sylow p -subgroup $\tau_p(A)$ of A . Its maximal divisible subgroup $\delta(\tau_p(A))$ is a direct sum of a number of quasi-cyclic groups of type Z_{p^∞} . The cardinality of the set of summands involved is an invariant of A , which we denote by $\beta(A, p)$. These invariants can be traced back to Ulm (see [9]); they were explicitly introduced in [18]. Finally, we recall that $a \in A$ has *infinite p -height* if the equation $x^{p^k} = a$ has a solution in A for every positive integer k . We are now in a position to formulate

Theorem A *Let A be a torsion, abelian group and suppose that $\tau_p(A)/\delta(\tau_p(A))$ has no elements of infinite p -height. Then A is discriminating if and only if $\alpha(A, p, k)$ is either zero or infinite for every prime p and every positive integer k and $\beta(A, p)$ is either zero or infinite for every prime p .*

1.3 Co-discriminating groups

Dually, we term a group H *co-discriminating* if an arbitrary family $\mathcal{S}$ of groups separates H if and only if $\mathcal{S}$ discriminates H . We will focus here on a large class of co-discriminating groups, *groups without zero divisors*, which were introduced in [5] in a rather different context. We recall that a group H has no zero divisors if given any non-trivial elements $a, b \in H$ there exists an element $x \in H$ such that $[a, b^x] \neq 1$. Groups without zero divisors are called *domains* and, as noted above (and easily proved directly from the definition) every such domain is co-discriminating. These domains include, in particular, the *CSA groups* introduced in [5]. We recall that a group G is a *CSA group* if every maximal abelian subgroup M of G is malnormal, i.e., $M^g \cap M = 1$ for every $g \in G - M$. The class of CSA-groups is quite substantial. It includes all abelian groups, all torsion-free hyperbolic groups [16] and all groups acting freely on Λ -trees [1] as well as many one-relator groups – a complete description of one-relator CSA-groups was obtained by D. Gildenhuys, O. Kharlampovich and A. Myasnikov in [10]. It turns out that many free products with amalgamation and many one-relator groups are domains. In particular, a free product $A * B$ is a domain unless both the groups A and B are of order 2. This allows one to view a result of B. Baumslag [2] that a free product $A * B$ is separated by a non-abelian free group if and only if both the groups A and B are discriminated by a non-abelian free group in a slightly different light. This theorem of Baumslag follows immediately from the remarks above. Indeed, more generally, the corresponding result holds when the non-abelian free group is replaced by an arbitrary torsion-free group G which separates its own free square $G * G$, for instance every non-abelian torsion-free hyperbolic group (see [4]).

The main result of the paper is

Theorem B *Every one-relator group with more than 2 generators is a domain, in particular it is co-discriminating.*

It follows, perhaps surprisingly (see Section 4.3, Theorem 6), that every

one-relator group which is separated by a free group is discriminated by a free group.

2 Discriminating abelian groups

2.1 Discriminating groups

Recall that a group G is *discriminating* if every group separated by G is discriminated by G .

First we record the following well-known lemma, the proof of which is left to the reader (see [8]).

Lemma 1 *A group H is separated by a family of groups $S = \{G_i \mid i \in I\}$ if and only if H is a subgroup of a cartesian product of groups from S .*

The following criterion is useful.

Proposition 1 *A group G is discriminating if and only if $G \times G$ is discriminated by G .*

Proof. Suppose that the group $G \times G$ is discriminated by G and that a group H is separated by G . Now by Lemma 1, H can be embedded in a cartesian power P of G . Let $h_1, \dots, h_n$ be a finite number of non-trivial elements of H . Then there exists a projection π of P onto a finite direct product $G^k = G \times \dots \times G$ of k copies of G such that $\pi(h_i) \neq 1$ for $i = 1, \dots, n$. Since G discriminates $G \times G$ it follows readily that G discriminates G^k . Thus 2 implies 1. The reverse implication is immediate.

We make use of this criterion repeatedly in what follows.

We gather together a number of results about discriminating groups. Throughout this section, p will denote a prime.

Proposition 2 *Let D be a discriminating group. Then the following conditions hold.*

1. *If H is a finite subgroup of $D \times D$, then there is a finite subgroup of D which is isomorphic to H .*
2. *If H is a finite subgroup of $D \times D$ and H is contained in a divisible subgroup (i.e., extraction of n -th roots is possible in that subgroup for every positive integer n) of $D \times D$, then there is a finite subgroup of D which is contained in a divisible subgroup and is isomorphic to H .*

3. If H is a finite subgroup of a Sylow p -subgroup (i.e., a subgroup in which every element has order a power of p and which is maximal subject to this condition) of $D \times D$, then there is a finite subgroup of a Sylow p -subgroup of D which contains a subgroup isomorphic to H .
4. The number of elements of a given finite order in D is either zero or infinite.
5. If D_i is discriminating for every $i \in I$, then the cartesian product $\prod_{i \in I} D_i$ and the restricted direct product $\prod_{i \in I} D_i$, are discriminating.

Proof. If D is discriminating, then for every finite subgroup K of $D \times D$ there exists a homomorphism $\phi : D \times D \longrightarrow D$ which is monic on K . If a finite subgroup K of $D \times D$ is contained in a divisible subgroup or a Sylow p -subgroup of $D \times D$ then its image under ϕ has the same property. These remarks are enough for the proof of 1, 2, 3 and 4.

We prove next that $D = \prod_{i \in I} D_i$ is discriminating – the proof that $\prod_{i \in I} D_i$ is discriminating is similar and is left to the reader. For each $i \in I$ we choose a discriminating family of homomorphisms Φ_i from $D_i \times D_i$ into D_i . For any “ I -sequence” of homomorphisms

$$\phi \in \prod_{i \in I} \Phi_i$$

we define a homomorphism $\lambda_\phi : D \times D \longrightarrow D$ as follows: if $f, h \in D$ then

$$\lambda_\phi(f, h)(i) = \phi_i(f(i), h(i)), \quad i \in I.$$

Clearly, λ_ϕ is a homomorphism of $D \times D$ into D . Now, if we have finitely many non-trivial elements $(f_1, h_1), \dots, (f_n, h_n) \in D \times D$, then we can find a finite subset I_0 of I such that for each of the elements (f_k, h_k) there exists an index $i \in I_0$ such that $(f_k(i), h_k(i)) \neq 1$ in $D_i \times D_i$. Since D_i is discriminating there exists a homomorphism $\phi_i \in \Phi_i$ discriminating all the elements $(f_k(i), h_k(i))$ for $i \in I_0$. If, for each $i \notin I_0$, we choose ϕ_i to be any element in Φ_i , we obtain an I -sequence $\phi \in \prod_{i \in I} \Phi_i$. Obviously, the homomorphism λ_ϕ discriminates all of the elements (f_k, h_k) , which proves that D discriminates $D \times D$.

2.2 Abelian groups

Next we focus on the problem as to which abelian groups are discriminating. We shall use additive notation throughout this subsection.

If A is an abelian group, then we denote the torsion subgroup of A by $\tau(A)$ and, for each prime p , we denote its unique Sylow p -subgroup by $\tau_p(A)$. Notice that $\tau(A)$ is the direct sum of its Sylow p -subgroups:

$$\tau(A) = \bigoplus_p \tau_p(A).$$

Finally, let $\delta(A)$ be the maximal divisible subgroup of A . A is termed *reduced* if $\delta(A) = 0$. Every divisible subgroup of an abelian group is a direct summand; so $\delta(A)$ is a direct summand of A and therefore we can express A in the form $A = B \oplus \delta(A)$. It follows that $B \cong A/\delta(A)$ is reduced. Notice also, that every divisible abelian p -group is a direct sum of groups of type $Z(p^\infty)$, i.e., additive copies of the multiplicative group of all p^n -th roots of unity. The following proposition then holds.

Proposition 3 *Let A be an abelian group.*

1. *If A is torsion-free abelian, then A is discriminating.*
2. *If A is discriminating, then so too is its torsion subgroup $\tau(A)$.*
3. *If A is a torsion group, then A is discriminating if and only if all of its Sylow p -subgroups $\tau_p(A)$ are discriminating.*
4. *If A splits over $\tau(A)$, then A is discriminating if and only if $\tau(A)$ is discriminating.*
5. *If A is a non-trivial divisible p -group, then A is discriminating if and only if it is a direct sum of infinitely many copies of $Z(p^\infty)$.*
6. *If A is discriminating then so too is $\delta(A)$.*

Proof.

1. Let $(a_1, b_1), \dots, (a_n, b_n)$ be non-trivial elements of $A \oplus A$. Since extraction of roots in a torsion-free abelian group is unique, whenever it is possible, there exists a positive integer k such that the homomorphism from $A \oplus A$ into A defined by

$$(a, b) \mapsto a + kb \quad (a, b \in A)$$

is monic on the set $\{(a_1, b_1), \dots, (a_n, b_n)\}$. This completes the proof.

2. It follows immediately from the fact that $A \oplus A$ is discriminated by A , that $\tau(A \oplus A) = \tau(A) \oplus \tau(A)$ is discriminated by $\tau(A)$ and hence that $\tau(A)$ is discriminating.

3. An analogous argument to the one above suffices to prove the given assertion.

4. This follows on combining 1 and 2.

5. If A is a finite direct sum of copies of $Z(p^\infty)$ then it has finitely many elements of order p , hence by Proposition 2 (4) A is not discriminating. If A is a direct sum of infinitely many copies of $Z(p^\infty)$ then $A \cong A \oplus A$ and therefore A is discriminating.

6. The image of a divisible subgroup under a homomorphism is again divisible. Since $A \oplus A$ is discriminated by A , it follows that $\delta(A) \oplus \delta(A)$ is discriminated by $\delta(A)$. Consequently $\delta(A)$ is discriminating.

Now a torsion abelian group of finite exponent is a direct sum of cyclic groups (see, e.g., I. Kaplansky [11]). The following lemma is an immediate consequence of this remark and the definition.

Lemma 2 *Let A be a torsion abelian group. Then $\alpha(A, p, k)$ is equal to the cardinality of the set of cyclic summands of order p^k in any decomposition of $A[p^k]$ into a direct sum of cyclic groups.*

We are now in a position to prove our main theorem about discriminating abelian groups.

Theorem A *Let A be a torsion, abelian group and suppose that $\tau_p(D)/\delta(\tau_p(D))$ has no elements of infinite p -height. Then A is discriminating if and only if $\alpha(D, p, k)$ is either zero or infinite for every prime p and every positive integer k and $\beta(D, p)$ is either zero or infinite for every prime p .*

Proof. By Proposition 3, (3), we can assume that A is a p -group.

We begin by proving that if $\alpha(A, p, k) = \alpha$ is non-zero, then it must be infinite. Suppose the contrary, i.e., for some k , α is finite and non-zero. Then if we decompose $A[p^k]$ into a direct sum of cyclic groups, there are exactly α cyclic summands $C(p^k)$ of order p^k . It follows that $A[p^k]$ and therefore A itself, does not contain any direct power of $C(p^k)$ with more than α summands. But $A \oplus A$ has a subgroup which is a direct power of 2α cyclic summands of order p^k . So, by Proposition 2, (1), A is not discriminating. Therefore, if $\alpha(A, p, k) = \alpha$ is non-zero, it must be infinite.

We observe next, that by Proposition 3, (5) and (6), $\beta = \beta(A, p)$ is either 0 or infinite. We consider first the case where $\beta = 0$. So A is a reduced, torsion abelian p -group, without any elements of infinite p -height. If $A = 0$, there is nothing to prove. So we assume that $A \neq 0$. It suffices then to prove that if H is any finite subgroup of $A \oplus A$, then there is a homomorphism of $A \oplus A$ into A which is monic on H . Now every finite subgroup of a p -group P without elements of infinite height can be embedded in a finite direct summand of P (see, e.g., [11]). Therefore we can assume that H is of the form $H = K \oplus K \leq D \oplus D$, where K is a direct summand of A , say $A = K \oplus L$. Suppose that we decompose K into a direct sum of cyclic groups and that a cyclic summand of maximal order has order p^j . Since $\alpha(A, p, j)$ is infinite, there are infinitely many elements of order p^j in L and hence subgroups of L which are arbitrarily large direct powers of $C(p^j)$. Hence there is a homomorphism μ of A into itself which is identical on L and which maps K monomorphically into L . Consider now the homomorphism ρ of $A \oplus A$ into A which maps $A \oplus \{0\}$ identically onto A and maps $\{0\} \oplus A$ into A according to μ . Then ρ maps H monomorphically into A – indeed it maps H onto $K \oplus K$, thought of now as a subgroup of A itself.

We are left with the case where β is infinite. Let H be a finite subgroup of $A \oplus A$. We can assume, without loss of generality, that $H = K \oplus K \leq A \oplus A$. If we now decompose A into a direct sum $A = B \oplus \delta(A)$, then again we can assume that $K = K_1 \oplus A_1 \leq E \oplus \delta(A)$. Since β is infinite, we can find a monomorphism μ of K into $\delta(A)$ such that $\mu(K) \cap D_1 = 0$. It follows that the homomorphism of $A \oplus A$ into A which maps $A \oplus \{0\}$ identically onto A and maps $\{0\} \oplus A$ into A according to μ , is monic on H . This completes the proof of the theorem.

The following examples provide answers to some of the natural questions that might arise in the readers minds.

Example 1 *Let A be the direct sum of a countably infinite number of copies of the groups Z_{p^∞} and a group of order p . Then A is discriminating, but A modulo its maximal divisible subgroup is not.*

Example 2 *Let p be a prime and let A be the unrestricted direct sum of the cyclic groups of order p^i ($i = 1, 2, \dots$). Then A is discriminating and therefore so too is its torsion subgroup T . Notice that T is not a direct sum of cyclic groups (see, e.g., [9]).*

Example 3 Let p be a prime. For a positive integer i denote by D_i the direct sum of a countably infinite number of copies of the cyclic group of order p^i . Then D_i is discriminating by Theorem A. Now, the unrestricted direct sum $A = \bigoplus_i D_i$ of the groups D_i ($i = 1, 2, \dots$) is discriminating by Proposition 2. Notice again, that the torsion subgroup of A is not a direct sum of cyclic groups.

3 Co-discriminating groups

3.1 Co-discrimination and n -separation

Recall that a group H is *co-discriminating* if every group G separating H discriminates H .

We shall have need here of a new notion.

Definition 3 Let H be a group, let $\mathcal{G}$ be a family of groups and let n be a positive integer. Then we say that $\mathcal{G}$ n -separates H if given any n non-trivial elements $h_1, \dots, h_n$ of H , there exists a group $G \in \mathcal{G}$ and a homomorphism ϕ of H into G such that $\phi(h_i) \neq 1$ for $i = 1, \dots, n$.

Notice, in particular, that if $\mathcal{G}$ n -separates H , then $\mathcal{G}$ certainly separates H . It turns out that, under the right circumstances, if $\mathcal{G}$ 2-separates H , then $\mathcal{G}$ actually discriminates H .

Some of the results that we will prove here are straightforward generalizations of some analogous results of B. Baumslag concerned with groups discriminated by free groups [2].

We recall first the

Definition 4 A group G is called *commutative transitive* if commutation is a transitive relation on the set of all non-trivial elements of G , i.e., if a commutes with b and b commutes with c , then a commutes with c , provided a, b, c are non-trivial.

It is not hard to see that CSA-groups are commutative transitive [16].

The following proposition is useful.

Proposition 4 Let H be a group and let $\mathcal{G}$ be a family of CSA-groups. Then the following hold:

1. Suppose that $\mathcal{G}$ 2-separates H . Then H is a CSA-group.
2. If H is commutative transitive and separated by $\mathcal{G}$, then H is a CSA-group.

Proof. 1. Suppose that H is not a CSA-group. Then there exists a maximal abelian subgroup M of H which is not malnormal. Hence there is a non-trivial element $a \in M$ and an element $x \notin M$ such that $a^x \neq a$ with $a^x = b \in M$. Put $c = [a, x]$. Notice that $c \neq 1$ and that $c \in M$. Now there exists a homomorphism ϕ of H into a group $G \in \mathcal{G}$ such that $\phi(a)$ and $\phi(c)$ are again non-trivial. Let M' be a maximal abelian subgroup of G containing $\phi(M)$. Since G is a CSA-group, $\phi(x) \in M'$. But then $[\phi(a), \phi(x)] = 1$ and therefore $\phi(c) = 1$, a contradiction. This completes the proof of 1.

2. If H is abelian, then H is a CSA-group. Assume then that H is non-abelian. It follows from the assumption that H is commutative transitive, that H has a trivial center. We will prove that $\mathcal{G}$ 2-separates H and then invoke 1. Suppose then that a and b are a pair of non-trivial elements of H . If there exists an element $x \in H$ such that $[a, b^x] \neq 1$, then there exists a homomorphism of H into a group in $\mathcal{G}$ such that the image of $[a, b^x]$ is not equal to 1. Consequently neither the image of a nor that of b is 1, as desired. On the other hand, suppose that a commutes with all of the conjugates of b in H . Then it follows from commutative transitivity that the normal closure of b in H is abelian, i.e., that H contains a non-trivial normal abelian subgroup. Hence we can find a homomorphism of H onto a non-abelian subgroup H' of a group $G \in \mathcal{G}$ which contains a non-trivial abelian normal subgroup M' . Let M be a maximal abelian subgroup of G containing M' . Then M is not malnormal in G , which is impossible since G is a CSA-group. This completes the proof.

Corollary 1 *Let G be a non-abelian CSA-group. Then $G \times G$ is separated by G but not discriminated by G .*

Proof. $G \times G$ is obviously separated by G . If G is non-abelian, then commutation in $G \times G$ is not transitive on the set of all non-trivial elements. Hence, $G \times G$ is not a CSA-group. Thence, by Proposition 4, $G \times G$ can not be discriminated by the CSA-group G . In particular, non-abelian CSA groups are not discriminating.

We remark, in passing, that during the course of the proof of Proposition 4, we have also proved that a non-abelian CSA-group G satisfies the following

property: if a and b are non-trivial elements of G , where we allow for the possibility that $a = b$, then there exists an element $x \in G$ such that $[a, b^x] \neq 1$. Such groups are called *groups without zero divisors* or, more frequently, *domains*.

Theorem 1 *Every group without zero divisors is co-discriminating.*

Proof. Let H be a group without zero divisors and let $h_1, \dots, h_n$ be finitely many, non-trivial elements in H . There exists an element $x_1 \in H$ such that $[h_1, h_2^{x_1}] \neq 1$. Hence there exists an element $x_2 \in H$ such that $[[h_1, h_2^{x_1}], h_3^{x_2}] \neq 1$ and so on. It follows that we can find elements $x_1, x_2, \dots, x_{n-1} \in H$ such that

$$c = [\dots [[h_1, h_2^{x_1}], h_3^{x_2}], \dots, h_n^{x_{n-1}}] \neq 1.$$

Now suppose that H is separated by a family of groups $\mathcal{G}$. Let ϕ be a homomorphism of H into a group $G \in \mathcal{G}$ such that $c^\phi \neq 1$. Then ϕ maps each of the elements $h_1, \dots, h_n$ non-trivially into G , as desired.

Corollary 2 *Every non-abelian CSA group is co-discriminating.*

4 Domains

4.1 Domains and products

Proposition 5 *Let A and B be non-trivial groups. Then $A * B$ is a domain unless both A and B are of order 2.*

Proof. Suppose that the order of B is at least 3. Let x and y be non-trivial elements of $A * B$. We claim that there are elements $g, h \in A * B$ such that $[x^g, y^h] \neq 1$. We can assume that x and y are cyclically reduced.

If $|x| = |y| = 1$ (here $|x|$ is the syllable length of x), then $[x, y] \neq 1$ provided x and y lie in different factors, and if they both lie in the same factor, say A , then $[x, y^b] \neq 1$ for any non-trivial $b \in B$.

Suppose now that $|x| \geq 2$. Then we can assume (conjugating if necessary) that $x = a_1 b_1 \dots a_n b_n$ ($1 \neq a_i \in A$, $1 \neq b_i \in B$). Again, conjugating if necessary, we can assume that y is a strictly alternating product of elements from A and B with the first and last elements coming out of B . In this event, the products xy and yx have different syllable length - so x and y do not commute, as desired.

If $|A| = |B| = 2$, then $A * B$ has a non-trivial normal abelian subgroup (the derived subgroup). In this event $A * B$ is not a domain.

Notice, that we do not require that the factors A and B in the lemma above be domains. Under the assumptions that A and B are domains, one can prove the following result (see [5] for details).

Theorem 2 *Let A and B be domains. Suppose that C is a subgroup of both A and B satisfying the following condition:*

$$\text{if } c \in C, c \neq 1, \text{ either } [c, A] \not\subseteq C \text{ or } [c, B] \not\subseteq C.$$

*Then the amalgamated free product $H = A *_C B$ is a domain.*

Now we are in the position to prove the following generalization of the result of B. Baumslag [2] referred to in the introduction.

Theorem 3 *Let A and B be non-trivial groups not both of order 2. Let G be a group which separates its own free square $G * G$ (in particular a non-abelian torsion-free hyperbolic group). Then the following conditions are equivalent:*

1. G separates $A * B$;
2. G discriminates $A * B$;
3. G discriminates each of A and B .

Proof. By Proposition 5 the group $A * B$ is a domain. So, by Theorem 1, 1 and 2 are equivalent.

3 immediately follows from 2 since A and B are subgroups of $A * B$.

Suppose now that both the groups A and B are discriminated by G . Then in the natural way $A * B$ is discriminated by $G * G$. It is then easy to see that if G separates $G * G$, then, by transitivity of separation, G separates $A * B$.

4.2 One-relator groups

The main theorem of this paper is the following

Theorem B *Let $G = \langle X \mid r = 1 \rangle$ be a one-relator group with $|X| > 2$. Then G is a domain.*

We shall make use of the following lemma.

Lemma 3 *Let N be a domain and let G be a semi-direct product of N and an abelian group. Then G is a domain if and only if G has trivial center.*

Proof. Clearly, if G is a domain then the center of G is trivial.

Suppose now that the center of G is trivial. Observe, that if x and y are a pair of zero divisors in G , then their normal closures $gp_G(x)$ and $gp_G(y)$ in G , commute:

$$[gp_G(x), gp_G(y)] = 1.$$

If $u, v \in G$, then

$$[x, u] \in gp_G(x), \quad [y, v] \in gp_G(y)$$

and hence

$$[gp_G([x, u]), gp_G([y, v])] = 1,$$

Now the center of G is trivial. Therefore there exist elements u and v in G such that $[x, u]$ and $[y, v]$ are non-trivial. But this implies that both $[x, u]$ and $[y, v]$ are zero-divisors in G . Since G/N is abelian, both $[x, u]$ and $[y, v]$ are contained in N . It follows that, in particular, $[x, u]$ is a zero-divisor in N , which contradicts the assumption that N is a domain.

We are now in a position to prove our main theorem. Suppose then that $X = \{t, b, c, \dots\}$ and that

$$G = \langle X; r = 1 \rangle, \tag{1}$$

where r is a cyclically reduced word in X . Let $\sigma_x(r)$ denote the sum of the exponents of all of the occurrences of x in r . We divide the proof into two cases.

Case 1: $\sigma_t(r) = 0$. Denote by N the normal closure in G of $X - t$. Put

$$b_i = t^i b t^{-i}, \quad c_i = t^i c t^{-i}, \dots,$$

Then N is generated by $Y = \{b_i, c_i, \dots\}$. Since $\sigma_t(r) = 0$, r can be re-expressed as a word $s = s(\dots, b_j, \dots, c_k, \dots)$ in the generators Y , where each of the generators x occurring in r is replaced by x_i , where the subscript i is the sum of the t -exponents of the subword of r preceding x . It follows then, that

$$t^i r t^{-i} = t^i s(\dots, b_j, \dots, c_k, \dots) t^{-i} = s(\dots, b_{j+i}, \dots, c_{k+i}, \dots), \quad (i \in \mathbb{Z}).$$

Put

$$s_i = s(\dots, b_{j+i}, \dots, c_{k+i}, \dots), \quad (i \in \mathbb{Z}).$$

Using the Reidemeister-Schreier method (see, for example [13]) it is not hard to prove that N has the following presentation:

$$N = \langle b_i, c_i, \dots \ (i \in Z) ; s_i = 1 \ (i \in Z) \rangle.$$

Let α and β be correspondingly the minimal and the maximal index of b that occurs in the word s . Now, for arbitrary non-negative integers i, j put

$$N_{-i,j} = \langle c_k, \dots \ (k \in Z), b_{\alpha-i}, b_{\alpha-i+1}, \dots, b_{\beta+j} \mid s_{-i} = 1, s_{-i+1} = 1, \dots, s_j = 1 \rangle.$$

It follows (see [13]) that N is the union of the following ascending chain of subgroups

$$N_{0,0} \leq N_{0,1} \leq N_{-1,1} \leq N_{-1,2} \dots \leq N_{-i+1,i} \leq N_{-i,i} \leq N_{-i,i+1} \leq \dots$$

Notice that in the presentation for $N_{-i,j}$ above, there are infinitely many generators and finitely many relators, so each $N_{-i,j}$ is a free product of two infinite groups. Hence, by Proposition 5, for each $i, j \geq 0$, the group $N_{-i,j}$ is a domain. Since the union of an ascending chain of domains is again a domain, we conclude that N is a domain. Observe that the group G is the semidirect product of N and the infinite cyclic group generated by t . K. Murasugi [15] has proved that every one-relator presentation with more than two generators, defines a group with trivial center. So by Lemma 3, G is a domain.

Case 2: $\sigma_x(r) \neq 0$, for every $x \in X$.

Denote by $F = F(X)$ a free group on X . Observe, that $r \notin [F, F]$. Consequently there exists a free basis $Y = \{y_1, y_2, \dots\}$ of F (see [13] Theorem 3.5), such that

$$r = y_1^e y'$$

where here e is a positive integer and y' is an element in the derived group of F , expressed in terms of the new basis Y of F . Clearly, $\sigma_y(s) = 0$ for every $y \in Y - \{y_1\}$. So if we present G on the generators Y , we find that

$$G = \langle Y; y_1^e y' \rangle.$$

So we are back to Case 1 and G is a domain, as before.

The condition $|X| \geq 3$ is essential, since, for example the following one-relator groups are not domains (they contain non-trivial abelian normal subgroups):

$$\langle x, y \mid x^k = y^\ell \rangle, \quad \langle x, y \mid y^{-1}xy = x^\ell \rangle, \quad ((k, \ell \geq 2).$$

It is unclear whether the only one-relator groups which are not domains are those which contain abelian normal subgroups, as in the examples above.

4.3 One-relator groups which are separated by free groups are discriminated by them

The object of this subsection is to prove the

Theorem 6 *Every one-relator group separated by a free group F is discriminated by the free group F .*

Since every one-relator group with at least three generators is, by Theorem B, a domain and hence, by Theorem 1, co-discriminating, it suffices to prove the following lemma.

Lemma 4 *Let G be a group generated by two elements. Then the following conditions are equivalent:*

1. G is separated by a free group F ;
2. G is discriminated by a free group F ;
3. G is either free or free abelian.

Proof. Notice, that 3 implies 2 (in the case when G is free abelian, it follows from Section 2 that G is discriminated by Z , hence by F). It is obvious that 2 implies 1. So it is enough to prove that 1 implies 3. Observe that every group separated by a torsion-free group is torsion-free; hence G is torsion-free. If G is abelian, then G is free abelian. Suppose now that G is generated by a and b and that $[a, b] \neq 1$. Then there exists a homomorphism $\phi : G \rightarrow F$ such that $[\phi(a), \phi(b)] \neq 1$. Therefore the subgroup generated by $\phi(a)$ and $\phi(b)$ is a free group of rank 2; hence ϕ is an isomorphism.

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